

# Compression-Aware Abstention: Teaching LLMs to Refuse When KV-Compression Masks Remove Answer Evidence

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## Abstract

KV-cache compression reduces LLM inference memory by evicting context tokens, but when the evicted tokens contain answer-bearing evidence, the model may hallucinate instead of recognizing that the compressed context is insufficient. We address this failure from a behavioral perspective: to our knowledge, this is the first work to formulate *compression-aware abstention* as a learning problem, in which a model learns to answer when supporting evidence survives compression and abstain when it does not. We construct supervision from compressor survival masks and tight answer-bearing spans, labeling examples as CONFIDENT when evidence survives and ABSTAIN when it is removed. A 10.1M-parameter LoRA adapter trained on  $\sim 2.6$ K MuSiQue 2-hop QA examples reduces base-model hallucinations by 97% under prompt-style truncation while preserving correct answering on evidence-retaining examples. Unlike prompt-only abstention baselines, which over-abstain on many answerable high-retention examples, the trained adapter learns a conditional policy. We also evaluate the method under actual compressed-cache decoding, where multi-compressor training yields a  $6\text{--}22\times$  relative lift over the unaided base on evidence-retaining examples. Controlled-deletion experiments show that the learned behavior is driven by evidence content rather than input length alone.

## 1 Introduction

KV-cache compression methods reduce inference memory by evicting context tokens deemed unimportant by some scoring rule (Li et al., 2024; Zhang et al., 2023; Kim et al., 2025; Devoto et al., 2025). They are designed to be *behavior-preserving*, on the assumption that low-importance tokens carry no answer-relevant signal. When the eviction rule drops a token that is answer-bearing for some downstream query, the model is left with an input missing exactly the evidence it needs, and tends to fab-

ricate a confident wrong answer rather than acknowledge the loss. The qualitative pattern is documented (Yang et al., 2024); existing remedies are representation-side (keep evicted entries in low-precision form rather than dropping them).

We propose a behavioral remedy: train the model to recognize when the surviving context cannot support an answer and respond honestly, answering when the supporting evidence survives and refusing when it does not. We refer to this as *compression-aware abstention*. To our knowledge, this is the first work to address KV-cache-compression-induced evidence loss from a behavioral perspective, by training the model to abstain when the surviving compressed context no longer supports an answer.

**Scope.** We construct training data from compression masks and feed the surviving tokens to the model as a normal prompt; we then test, separately, whether the trained behavior survives when the model decodes from a physically compressed cache (KVzipPress at inference). §5.5 reports both inference paths: the prompt-style (ps) path used to build training data and the compressed-cache (cc) path that matches deployment. Our objective is complementary to training-aware compression such as KV-Distill (Chari et al., 2025), which instead preserves uncompressed-model behavior under compression (§6).

**Contributions.** We make five contributions:

**C1.** We formulate compression-aware abstention as a behavioral learning problem. Rather than only asking whether KV-cache compression preserves the uncompressed model’s answer, we ask whether the compressed context still contains enough evidence to support an answer. We define examples as CONFIDENT when answer-bearing evidence survives compression and ABSTAIN when it does not, and train the model to answer or refuse accordingly.

**C2.** We show that tight evidence spans are crucial for making this learning problem well-posed. Paragraph-level support labels wash out the effect of compression because long support paragraphs often remain partially present even when key evidence is lost. By labeling survival over short answer-bearing spans, we obtain the per-example variation needed to distinguish answerable from unanswerable compressed contexts.

**C3.** We show that a small adapter can sharply reduce compression-induced hallucination while preserving answerability. A 10.1M-parameter LoRA adapter trained on approximately 2.6K compression-labeled MuSiQue examples eliminates 97% of base-model hallucinations under prompt-style truncation, while preserving correct answering on CONFIDENT-gold examples.

**C4.** We show that learned abstention is more conditional than prompt-only abstention and more robust when trained across compressors. Prompt-only baselines tend to over-abstain when evidence is actually present: at high retention, they abstain on about half of CONFIDENT-gold examples, while the trained adapter abstains on only 6–10%. Training on multiple compressor masks further improves transfer beyond a single compression pattern.

**C5.** We evaluate the method under deployment-relevant compressed-cache decoding and test what the adapter has learned. Under actual compressed-cache inference, the unaided base model nearly collapses on evidence-retaining examples; multi-compressor training improves correct answering from 4% to 25% aggregate and from 1.4% to 32% at high retention. Controlled-deletion experiments show that the adapter’s decisions depend on which evidence tokens survive, not merely on input length.

## 2 Compression-aware abstention as a learning problem

We give a formal definition of the learning problem the rest of the paper addresses. The definition makes three things precise: (i) what a compression-truncated input is, (ii) how the gold label is determined from the survival mask and the supporting-span structure, and (iii) what the trained adapter optimizes.

**A motivating example.** Consider the 2-hop question “Who founded the company that distributed the film UHF?”, whose answer is *Mike Medavoy*. Deriving this answer from the context requires one

short, specific piece of evidence (the phrase naming the founder) to survive compression. Now suppose a KV-cache compressor keeps most of the surrounding context (the film, its release, other people and companies) but, scoring that particular name as low-importance, evicts the few tokens that spell it out. The truncated context still reads fluently and still looks on-topic, so a base model prompted with it tends to supply a confident but unsupported name (in our data the base answers “Kevin Smith”). The correct behavior depends entirely on whether that one phrase survived: if the founder’s name is kept, the example is CONFIDENT and the model should answer *Mike Medavoy*; if it is evicted, the example is ABSTAIN and the model should refuse, because the surviving text no longer determines the answer. The rest of this section makes “a supporting phrase survives” quantitative (Eq. 1) and turns this all-or-nothing flip into a training label (Eq. 2).

**Compression-truncated input.** Let  $c \in \mathcal{V}^{n_c}$  be a context tokenized into  $n_c$  tokens, and let  $q$  be a query. A KV cache compressor at retention ratio  $r$  produces a per-token *survival mask*  $\mathbf{m} \in \{0, 1\}^{n_c}$  with  $\frac{1}{n_c} \sum_i m_i \approx r$ , and the associated compression-truncated input  $\tilde{c}(\mathbf{m}) = (c_i : m_i = 1)$  concatenates the surviving tokens. Different compressors select *which* tokens survive; the model is then run on  $\tilde{c}$  as a standard prompt.

**Evidence survival.** Let  $S = \{S_1, \dots, S_K\}$  be a set of *supporting spans*: contiguous token ranges in  $c$  that are jointly required for the answer  $a$  to be derivable from the context. Define the *evidence-survival fraction* of span  $S_k$  under mask  $\mathbf{m}$  as

$$\rho(\mathbf{m}, S_k) = \frac{\sum_{i \in S_k} m_i}{|S_k|}. \quad (1)$$

The granularity of  $S_k$  is decisive (§3.1): when  $|S_k|$  is large,  $\rho$  averages out and per-example label variance vanishes; when  $|S_k|$  is the size of the verbatim answer string (median 3 tokens),  $\rho$  is informative.

**Gold label.** Given thresholds  $\tau_{\text{low}} < \tau_{\text{high}} \in [0, 1]$ , the gold label  $y(\mathbf{m}, S)$  is

$$y = \begin{cases} \text{CONFIDENT} & \text{if } \min_k \rho(\mathbf{m}, S_k) \geq \tau_{\text{high}} \\ \text{ABSTAIN} & \text{if } \min_k \rho(\mathbf{m}, S_k) < \tau_{\text{low}} \\ \text{DROP} & \text{otherwise (excluded).} \end{cases} \quad (2)$$

Examples whose minimum span survival lands in the uncertain zone  $[\tau_{\text{low}}, \tau_{\text{high}})$  are dropped from training to avoid noisy targets.

**Target string and learning objective.** Each  $(c, q, a, \mathbf{m}, S)$  tuple is mapped to a target string  $t = a$  if  $y = \text{CONFIDENT}$  and  $t = t_\perp$  if  $y = \text{ABSTAIN}$ , where  $t_\perp$  is a fixed refusal string. The training set  $\mathcal{D}$  consists of  $(\mathbf{m}, q, t)$  triples for which  $y \neq \text{DROP}$ , with truncated input  $\tilde{c}(\mathbf{m})$ . The adapter parameters  $\theta$  are LoRA weights (Hu et al., 2022) attached to a frozen base LLM  $f$ , optimizing

$$\mathcal{L}(\theta) = - \sum_{(\mathbf{m}, q, t) \in \mathcal{D}} \log f_\theta(t \mid \tilde{c}(\mathbf{m}), q). \quad (3)$$

This loss splits over the two label classes,  $\mathcal{L}(\theta) = \mathcal{L}_{\text{CONF}}(\theta) + \mathcal{L}_{\text{ABS}}(\theta)$ , summing the cross-entropy over  $\mathcal{D}_{\text{CONF}}$  (target  $a$ ) and  $\mathcal{D}_{\text{ABS}}$  (target  $t_\perp$ ). The two terms incentivize opposite per-example behaviors:  $\mathcal{L}_{\text{CONF}}$  penalizes refusing when the truncated input does support the answer,  $\mathcal{L}_{\text{ABS}}$  penalizes answering when it does not. Both are needed for calibrated bidirectional behavior: optimizing only  $\mathcal{L}_{\text{ABS}}$  yields a fixed abstention prior (the prompt-only failure mode of §5.2), while optimizing only  $\mathcal{L}_{\text{CONF}}$  approaches a preservation-style objective (Chari et al., 2025) (always answer). We offer this as the natural reading of Eq. 3, not as an ablated claim.

**Class balance and comparison.** Because the label distribution is U-shaped in  $r$  (mostly ABSTAIN at low retention, mostly CONFIDENT at high), the mid-retention regime ( $r \approx 0.50$ ) is where both classes co-occur and Eq. 2 is informative per-example; we apply a per-(compressor,  $r$ ) class-balance floor at  $r \geq 0.20$  so both loss terms see comparable counts (§3.2, Appendix E). The objective is complementary to preservation-style training (Chari et al., 2025): once  $\rho(\mathbf{m}, S) < \tau_{\text{low}}$ , the uncompressed-model answer is no longer a meaningful target, and the right target is  $t_\perp$ .

### 3 Method

#### 3.1 Tight-span localization

Eq. 1 is informative only when  $|S_k|$  is small. For each MuSiQue (Trivedi et al., 2022) 2-hop example, the `question_decomposition` field gives each single-hop sub-answer as a short atomic string (e.g., “Mike Medavoy”); we locate each verbatim in its supporting paragraph, yielding a short span (median 3 tokens, 99th percentile 11; located in 100% of our pool).

Tight spans are what make the labeling discriminative. With full supporting paragraphs ( $\sim 100$

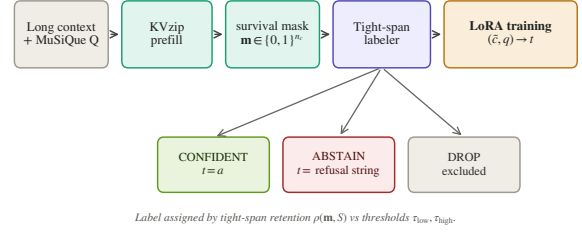


Figure 1: Training pipeline. KVzip prefill produces a per-token survival mask  $\mathbf{m}$  at the target retention ratio. The labeler computes evidence-survival  $\rho(\mathbf{m}, S_k)$  (Eq. 1) over tight supporting spans and assigns CONFIDENT, ABSTAIN, or DROP via Eq. 2. The LoRA adapter is trained on (truncated input  $\tilde{c}$ , query  $q$ )  $\rightarrow$  target  $t$ .

tokens) as  $S_k$ ,  $\rho$  averages out: at  $r = 0.50$  every paragraph keeps roughly half its tokens, so every example falls in the uncertain zone and is dropped (label entropy 0.000, vs 0.95 for tight spans; Appendix A). The pivot to verbatim answer spans is the reason the rest of the paper has signal to fit.

#### 3.2 Pipeline and adapter

Figure 1 summarizes the construction pipeline. The compressor produces  $\mathbf{m}$ ; the labeler computes  $\rho(\mathbf{m}, S_k)$  for each tight span and assigns  $y$  via Eq. 2 with thresholds  $\tau_{\text{low}} = 0.3$ ,  $\tau_{\text{high}} = 0.8$  (set a priori). The resulting (truncated input, query, target) triple enters  $\mathcal{D}$  if  $y \neq \text{DROP}$  (per-ratio label entropy in Appendix A).

We attach a 10.1M-parameter LoRA adapter (Hu et al., 2022) (rank 16) to the four attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`; MLP modules unadapted), 0.13% of Qwen2.5-7B-Instruct’s parameters. Optimizer, learning-rate sweep, and precision settings are in Appendix R.

Train  $\approx 2.6\text{K}$  instances (30/70 minority-class floor); validation 463 at the natural class distribution, disjoint by example ID. We call this KVzip-only adapter *v1*; §5.3 introduces a multi-compressor variant, *mix*, and §5.5 distinguishes the prompt-style and compressed-cache versions of each (*v1-ps*, *mix-ps*, and *mix-cc*).

### 4 Experimental setup

**Models.** Qwen2.5-7B-Instruct (Qwen Team, 2024) for all main results; Llama-3.1-8B-Instruct (Llama Team, Meta AI, 2024) as a held-out base model.

**Compressors.** KVzip (Kim et al., 2025) for training masks, via the authors’ reference implementa-

| $r$         | $n$       | C/A          | base        | adp         | $\Delta\text{hon}$ |
|-------------|-----------|--------------|-------------|-------------|--------------------|
| 0.05        | 100       | 0/100        | 0.15        | 1.00        | +0.85              |
| 0.10        | 95        | 1/94         | 0.27        | 0.99        | +0.72              |
| 0.20        | 82        | 2/80         | 0.30        | 0.92        | +0.62              |
| 0.30        | 65        | 3/62         | 0.35        | 0.79        | +0.44              |
| <b>0.50</b> | <b>48</b> | <b>22/26</b> | <b>0.44</b> | <b>0.66</b> | <b>+0.22</b>       |
| 0.80        | 73        | 69/4         | 0.51        | 0.60        | +0.09              |
| <i>agg.</i> | 463       | 97/366       | 0.32        | 0.86        | +0.54              |

Table 1: Per-ratio honest accuracy on Qwen val. Mean over three seeds (42, 43, 44). The 0.50 row is the only ratio at which the gold class is balanced and at which a model conditioning on  $\tilde{c}$  can do better than a fixed prior (§5.2).

tion: `kvpress`’s `KVzipPress` defers eviction in a way incompatible with prompt-style mask extraction, though §5.5 uses it for compressed-cache inference. Expected Attention (Devoto et al., 2025) and SnapKV (context-only and native query-aware modes) (Li et al., 2024) are held-out compressors, accessed via the `kvpress` library (NVIDIA, 2026). All are calibrated to retain an identical number of context tokens at each  $r$ .

**Retention ratios.**  $r \in \{0.05, 0.10, 0.20, 0.30, 0.50, 0.80\}$ . The gold class distribution is U-shaped in  $r$  (Table 1): almost all ABSTAIN at low retention, almost all CONFIDENT at high, and balanced at the headline mid-slice  $r = 0.50$ .

**Outputs and metric.** Greedy decoding, max 64 new tokens. Outputs are classified into six categories (Appendix B). *Honest accuracy* is the fraction of outcomes that are either CONFIDENT-gold with the model emitting the answer  $a$ , or ABSTAIN-gold with the model emitting the refusal  $t_\perp$ .

**Compute and reproducibility.** All training and evaluation run on a single NVIDIA H100 80GB GPU. Code, evaluation outputs, adapters, datasets, and a script that recomputes every reported number are released.<sup>1</sup>

## 5 Results

### 5.1 Headline

Training improves honest accuracy at every retention ratio (Table 1, Figure 2), and the gain is driven by a collapse in hallucination: confident wrong answers on ABSTAIN-gold (the *hallucinated* category of Appendix B) fall from 232 to 7. The adapter’s abstention rate falls monotonically as retention rises,

<sup>1</sup><https://github.com/mali-kh/compression-aware-abstention>

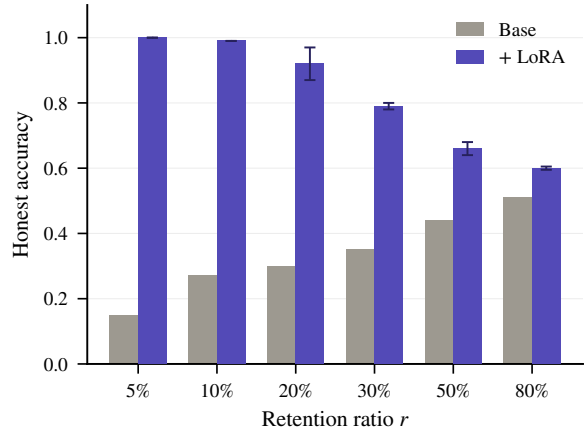


Figure 2: Per-ratio honest accuracy on Qwen val (KVzip masks). Bars are means across three seeds; error bars are seed spread. The adapter improves honest accuracy at every retention ratio.

consistent with evidence-sensitive behavior, though this alone does not rule out a retention-ratio heuristic, which §5.2 and §5.6 test directly. Conditional answer quality is preserved (Appendix C).

**No over-abstention cost.** A refusal-biased adapter could cut hallucinations on ABSTAIN-gold while also refusing answerable CONFIDENT-gold examples. It does not: the adapter’s correct-answer rate on CONFIDENT-gold is at least the base’s in all eight prompt-style cells across four compressors (*answerability preservation* ratio 1.00–1.36; on KVzip, 47/97 for base vs 54/97 for v1 and 59/97 for the mixture, with Wilson 95% CIs in Appendix L).

### 5.2 Baselines

Aggregate honest accuracy is useful for orientation, but it is not a reliable operationalization of compression-aware abstention by itself. The reason is that the gold labels are highly skewed as a function of the retention ratio  $r$ : at low  $r$ , almost all examples are ABSTAIN-gold, while at high  $r$ , almost all are CONFIDENT-gold. A trivial policy that never sees the truncated input  $\tilde{c}$  can exploit this structure: predict ABSTAIN when  $r < \tau$  and CONFIDENT otherwise. Sweeping  $\tau$  on Qwen val, the best threshold ( $\tau = 0.80$ ) reaches aggregate honest accuracy 0.93, exceeding both v1 (0.84) and the mixture adapter (0.86). This does not make the threshold policy compression-aware; it only shows that aggregate accuracy rewards the U-shaped class distribution. We therefore use the diagnostic slices below to ask whether a method is actually sensitive



| Compressor          | gold (C/A)  | best- $\tau$ | P2          | Frag        | v1          | Mix         |
|---------------------|-------------|--------------|-------------|-------------|-------------|-------------|
| KVzip               | 22/26       | 0.54         | 0.54        | 0.44        | 0.65        | <b>0.69</b> |
| ExpAttn             | 23/19       | 0.55         | 0.60        | 0.38        | 0.67        | <b>0.71</b> |
| SnapKV (ctx)        | 18/21       | 0.54         | 0.64        | 0.33        | 0.46        | <b>0.64</b> |
| <i>SnapKV (nat)</i> | <i>28/9</i> | <i>0.76</i>  | <i>0.54</i> | <i>0.38</i> | <i>0.41</i> | <i>0.51</i> |

Table 2: Honest accuracy at  $r = 0.50$  across compressors. *v1*: adapter trained on KVzip masks only. *Mix*: adapter trained on KVzip + ExpAttn + SnapKV masks jointly. SnapKV-native at this slice has degenerate gold mix (28/9) and is not interpretable for this comparison; reported in italics for completeness.

to the evidence that survives in  $\tilde{c}$ .

**Two diagnostic slices.** The balanced mid-retention slice and the high-retention slice probe complementary failure modes. At  $r = 0.50$ , the gold labels are nearly balanced (22 CONFIDENT, 26 ABSTAIN), so a retention-ratio threshold is forced into one fixed prediction across the slice. This slice tests whether a method can distinguish answerable from unanswerable truncated inputs on a per-example basis. At  $r = 0.80$ , the gold labels are dominantly CONFIDENT; this slice tests the opposite failure mode, namely whether an abstention method over-refuses even when evidence usually survives. Together, these slices separate genuine evidence sensitivity from fixed abstention or fixed-answering priors.

**Baseline definitions.** *P2 prompt*: a light-touch abstention instruction prepended to the user message: “If the available context does not contain enough evidence to answer the question, respond with: ‘I cannot determine the answer to this question from the available context.’ Otherwise, answer the question.” No LoRA adapter is applied. *Fragment*: the same compression-truncated input  $\tilde{c}$  is presented to a clean base model, with no LoRA adapter and no abstention instruction. The Fragment baseline is input-matched to our adapter; the only difference is the learned LoRA weights. *best- $\tau$* : the best ratio-only threshold policy, with  $\tau = 0.80$  tuned on KVzip aggregate.

**Balanced slice: training helps; prompt inconclusive.** Table 2 shows that the mixture adapter improves over the Fragment baseline by 25–33pp at  $r = 0.50$  on the three interpretable compressor settings (per-compressor paired McNemar exact,  $p \in [0.0005, 0.012]$  on KVzip / ExpAttn / SnapKV-context; not significant on SnapKV-native). This establishes that the behavior is not obtained merely

by showing the base model the surviving text.

The comparison to the prompt-only baseline is more nuanced at this slice. Point estimates favor the mixture adapter on KVzip (+0.15) and ExpAttn (+0.12), tie on SnapKV-context ( $\pm 0$ ), and slightly favor P2 on SnapKV-native ( $-0.03$ ); none reaches per-compressor significance at  $n \in [37, 48]$  (per-compressor exact  $p$ -values and paired-bootstrap CIs in Appendix K). Pooling the four compressors into a 166-example paired test yields  $\Delta = +0.07$  ( $p = 0.17$ ). Thus the mid-slice suggests a positive trend over prompting, but the available sample does not make that comparison decisive. The cleaner prompt-vs-training result appears at high retention, where the same fixed-prior failure has a larger and more measurable effect.

**Fixed prior vs. learned policy.** At  $r = 0.80$ , where roughly 95% of the gold labels are CONFIDENT, the prompt-only baseline exposes its main failure mode: it over-abstains even when evidence usually survives. On KVzip, P2 honest accuracy drops to 0.34, below the base model’s 0.51. It abstains on 49% of the 69 CONFIDENT-gold examples, compared with 10% for the mixture adapter and 3% for v1. The pattern is uniform across compressors (Figure 3): P2 over-abstains on 49–51% of CONFIDENT-gold examples, while the mixture adapter over-abstains on only 6–10%.

Restricting to the CONFIDENT-gold subset at  $r = 0.80$ , the mixture adapter outperforms P2 on every compressor, with  $\Delta \in [+0.28, +0.34]$  and per-compressor paired McNemar exact  $p \leq 2 \times 10^{-4}$ . Pooled across the 260 CONFIDENT-gold examples at this retention, the advantage is  $\Delta = +0.32$  (95% paired-bootstrap CI  $[+0.25, +0.38]$ , McNemar  $p \approx 1.5 \times 10^{-16}$ ). This is the clearest baseline evidence in our setting: an abstention prompt tends to create a broad refusal prior, whereas the trained adapter learns a bidirectional policy that answers when evidence survives and abstains when it does not. This is precisely the behavior encouraged by the loss decomposition in Eq. 3 into  $\mathcal{L}_{\text{CONF}}$  and  $\mathcal{L}_{\text{ABS}}$ . Whether richer prompting strategies (for example, self-consistency over abstention judgments, chain-of-thought introspection, or verifier-style confidence checks) could approximate this behavior at inference time remains open; we do not test them here.

**Surviving text is not enough.** Finally, the Fragment baseline shows that correct abstention is a learned property of the adapter, not an automatic

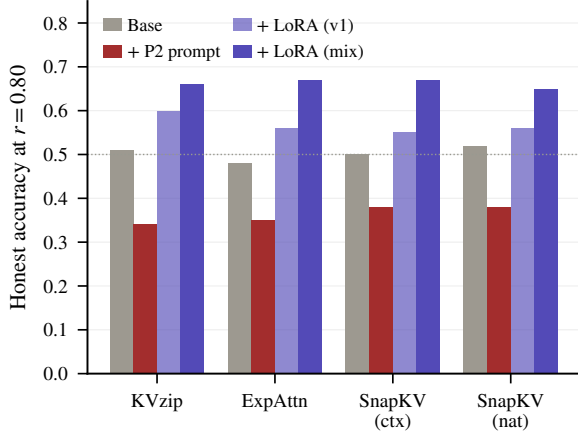


Figure 3: Honest accuracy at  $r = 0.80$  (high retention, dominantly CONFIDENT-gold) across four compressors. The P2 abstention prompt drops below base on every compressor (dotted line at base level); the trained adapter does not. Calibrated bidirectional behavior (abstain when evidence is missing *and* answer when it survives) requires per-example sensitivity to  $\tilde{c}$ , which the loss decomposition in Eq. 3 explicitly captures (§2).

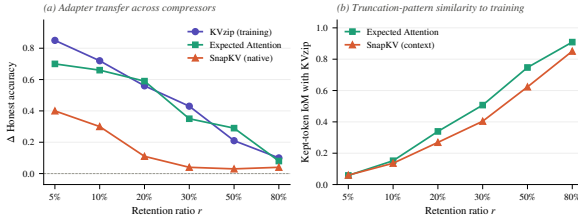


Figure 4: (a) Per-ratio  $\Delta$ honest accuracy of the v1 adapter (KVzip-trained) on three held-out compressors. Within-family transfer is clean (KVzip  $\rightarrow$  ExpAttn, aggregate +0.49 vs +0.53); cross-family transfer is partial (KVzip  $\rightarrow$  SnapKV-native, +0.20). (b) Differential transfer tracks kept-token intersection-over-min between each compressor and KVzip.

consequence of seeing a shortened context. Of the 89 examples where v1 correctly abstained on KVzip but hallucinated on SnapKV (the cross-compressor failure mode of §5.3), the Fragment baseline catches only  $13/89 = 15\%$  on the KVzip side: it hallucinates on 80% of the same examples the adapter abstains on.

### 5.3 Cross-compressor transfer and mixture training

A compression-aware abstention policy should not depend too tightly on the idiosyncrasies of one compressor’s mask. To test this, we first train v1 only on KVzip masks and evaluate it, without re-training, on three held-out compressors. The result is mixed. Transfer to Expected Attention is strong:

| Compressor          | v1    | Mix   | mix – v1 | Rescue       |
|---------------------|-------|-------|----------|--------------|
| KVzip               | +0.53 | +0.55 | +0.02    | —            |
| Expected Attn       | +0.49 | +0.57 | +0.08    | —            |
| <b>SnapKV (ctx)</b> | +0.21 | +0.53 | +0.32    | <b>82/89</b> |
| <b>SnapKV (nat)</b> | +0.20 | +0.43 | +0.23    | —            |

Table 3: Aggregate  $\Delta$ honest accuracy: v1 (KVzip-only) vs mixture training (KVzip + ExpAttn + SnapKV-context). *v1*: seed 42; *Mix*: mean across seeds 42 and 43. *Rescue*: of 89 v1-KVzip-correct  $\rightarrow$  v1-SnapKV-hallucinated examples, fraction the mixture catches under SnapKV evaluation (seed 42). Per-cell variance in Appendix E.

the aggregate gain remains close to the KVzip setting (+0.49 vs +0.53). Transfer to SnapKV is much weaker, with aggregate gains of only +0.21 for SnapKV-context and +0.20 for SnapKV-native (Figure 4a).

This pattern suggests that the adapter has learned two different behaviors. One is compressor-agnostic: when the surviving text contains enough evidence, the model can read and answer. The other is more compressor-specific: recognizing that evidence is missing depends partly on the deletion patterns seen during training. The transfer results track this interpretation. At  $r = 0.50$ , KVzip and Expected Attention agree on 75% of kept tokens, while KVzip and SnapKV agree on only 62%; the drop in transfer follows the same ordering (Figure 4b). Appendix D gives a qualitative decomposition of these “read-and-answer” and “recognize-missing-evidence” behaviors.

We therefore retrain on a mixture of masks from KVzip, Expected Attention, and SnapKV-context. The mixture set contains 8772 training instances, with per-(compressor,  $r$ ) class balance for  $r \geq 0.20$  and the natural class distribution at the extreme low-retention ratios. Hyperparameters match v1 except for  $\text{lr} = 2 \times 10^{-4}$ ; results are averaged over two seeds.

Mixture training largely closes the cross-compressor gap without sacrificing performance on the original compressor (Table 3). On KVzip, the aggregate gain is essentially unchanged (+0.53  $\rightarrow$  +0.55). On Expected Attention, it improves from +0.49 to +0.57. The largest gains appear on SnapKV, where v1’s partial transfer is the main failure mode: SnapKV-context improves from +0.21 to +0.53, and SnapKV-native from +0.20 to +0.43. The rescue analysis makes the same point at the example level: of the 89 cases where v1 abstained correctly under KVzip but hallucinated

under SnapKV, mixture training catches 82 under SnapKV evaluation.

Mixture training also improves calibration. The conditional wrong-when-answering rate decreases on 7 of 8 conclusive cells (Appendix C), and the KVzip-specific overcaution seen in v1 at  $r = 0.50$  disappears: mixture overcaution is 23%, matching the base model. Thus the benefit of mixture training is not merely higher honest accuracy; it makes the learned abstention policy less tied to a single compressor’s mask geometry.

#### 5.4 What is the load-bearing training signal?

Retraining v1 with masks drawn uniformly at random (at matched per-example retention counts) matches the mixture adapter on KVzip val and beats KVzip-only v1 on cross-family transfer. Under prompt-style inference, the load-bearing signal is therefore the abstention objective itself, not the importance-weighted structure of the compressor mask; single-compressor training additionally induces an anti-transfer signature. This motivates compressor-mixture training (§5.3), which §5.5 extends to compressed-cache inference. Full numbers are in Appendix P.

#### 5.5 Compressed-cache inference and cc-training

The results so far use prompt-style inference: we apply the compressor’s survival mask, feed the surviving tokens back to the model as an ordinary prompt, and measure whether the adapter answers or abstains. This is useful for controlled training and analysis, but it is not the deployment path. In actual compressed-cache inference, the model first prefills on the full context, the compressor evicts low-scoring K/V pairs from the cache, and generation proceeds from the compressed cache state. We therefore ask whether the learned abstention behavior survives this more realistic setting. We evaluate on the same Qwen validation set using kvpress’s KVzipPress at inference.

Compressed-cache decoding is substantially harder for the unaided base model. Under KVzipPress, base-cc reaches aggregate honest accuracy 0.186, which is 13pp below its prompt-style counterpart. The collapse is most visible on CONFIDENT-gold examples, where the correct-answer rate is only 4/97 aggregate and 1/69 at  $r = 0.80$ . In other words, when the answer-bearing evidence is labeled as surviving, the instruction-tuned base

| $r$         | base-cc | v1-ps cc     | mix-ps cc    | mix-cc cc    |
|-------------|---------|--------------|--------------|--------------|
| 0.05        | —       | 0.560        | 1.000        | <b>1.000</b> |
| 0.10        | —       | 0.495        | 0.958        | <b>0.989</b> |
| 0.20        | —       | 0.415        | 0.829        | <b>0.963</b> |
| 0.30        | —       | 0.323        | 0.600        | <b>0.923</b> |
| <b>0.50</b> | —       | <b>0.271</b> | <b>0.354</b> | <b>0.500</b> |
| 0.80        | —       | 0.233        | 0.315        | 0.315        |
| agg.        | 0.186   | 0.406        | 0.730        | 0.821        |

Table 4: Per-ratio honest accuracy under *actual* compressed-cache inference (KVzipPress) on Qwen val ( $n = 463$ ). *base-cc*: unmodified Qwen2.5-7B-Instruct under cc inference (per-ratio omitted; aggregate 0.186). *v1-ps*: prompt-style-trained, KVzip-only. *mix-ps*: prompt-style-trained, multi-compressor. *mix-cc*: compressed-cache-trained, multi-compressor (seed 42; per-seed values and two-seed mean in Appendix M).

model still often fails to use it once generation proceeds from the compressed cache.

The trained adapters recover much of this loss. Prompt-style multi-compressor training transfers surprisingly well to compressed-cache inference: mix-ps reaches aggregate honest accuracy 0.730, a +0.544 gain over base-cc. By contrast, the KVzip-only prompt-style adapter transfers much less well: v1-ps reaches 0.406, only +0.220 over base-cc. This mirrors the cross-compressor pattern in §5.3: training on a single mask family leaves the adapter more brittle when the inference-time corruption pattern changes.

Training directly in the compressed-cache regime adds another gain. Using the same multi-compressor mixture but computing the loss after cache eviction, mix-cc reaches aggregate honest accuracy 0.821, improving by +0.091 over mix-ps under the same cc inference path. The largest gains are at the intermediate retention ratios, especially  $r = 0.30$  and  $r = 0.50$ , where mix-cc improves over mix-ps by +0.32 and +0.15, respectively. At  $r = 0.80$ , mix-cc and mix-ps tie in honest accuracy (0.315 vs 0.315; McNemar on CONFIDENT-gold,  $p = 0.80$ ), indicating that high-retention compressed-cache answering remains the hardest case.

The answerability numbers clarify the scale of the recovery. On CONFIDENT-gold examples, multi-compressor cc-training improves correct answering from 4/97 to 24/97 aggregate, and from 1/69 to 22/69 at  $r = 0.80$ . This corresponds to a 6–22 $\times$  relative lift over base-cc, with non-overlapping Wilson 95% confidence intervals (Appendix L). The gain is large because base-cc nearly

collapses in this regime; the absolute rates also show that cc inference remains more difficult than prompt-style truncation.

Multi-compressor training is important even when training directly under compressed-cache decoding. A single-compressor cc-trained adapter ties mix-cc on aggregate honest accuracy, but at  $r = 0.80$  it produces a 61% wrong-answer rate on CONFIDENT-gold examples, compared with 30% for mix-cc (Appendix M). Thus the mixture is not merely improving the aggregate score; it keeps the answer-vs-abstain tradeoff better calibrated under the deployment-relevant inference path.

### 5.6 Mechanism: evidence content or input length?

The adapter’s decisions track *which* evidence survives, not how much: it responds to evidence content, not input length. This matters because the adapter sees a shortened input and could in principle be abstaining on length alone. Three controlled-deletion tests, all holding input length fixed, rule that out (Appendix O). The decisive one constructs, for each CONFIDENT-gold example at  $r = 0.50$ , two *equal-length* masks: one keeping KVzip’s highest-priority half, one its lowest-priority half. The adapter answers on the high-priority masks and abstains on the low-priority ones, where neither adapter answers a single example correctly and mix-ps abstains on all 22 with *zero* confident wrong answers; replacing only the answer-bearing tokens with random kept tokens raises abstention just as sharply. A length prior predicts identical behavior across these matched-length conditions; we observe the opposite.

### 5.7 Long-context generalization and base-model replication

The learned behavior generalizes beyond the training setup. On RULER multi-key NIAH (Hsieh et al., 2024) at context lengths up to 16K (four times the training maximum), the adapter preserves the same broadly declining-with- $r$  abstention shape seen on MuSiQue val and cuts aggregate hallucination rate by 36–43pp at every length (Appendix Q). Rerunning the full pipeline on Llama-3.1-8B-Instruct as a held-out base (independent training) reproduces the effect:  $\Delta$ honest accuracy +0.64 and 96% hallucination reduction, with the declining abstention curve preserved on both bases (Appendix G).

## 6 Related work

**KV-cache compression.** Algorithms for evicting low-importance KV pairs score importance variously by trailing-window attention (Li et al., 2024), accumulated past-attention with recency (Zhang et al., 2023; Liu et al., 2023), attention-sink heuristics (Xiao et al., 2024), layer-wise pyramidal allocation (Cai et al., 2024), context reconstruction (Kim et al., 2025), future-query attention (Devoto et al., 2025), or per-head retrieval/streaming roles (Xiao et al., 2025). Others reduce cache cost without evicting, through query-aware page selection (Tang et al., 2024) or low-bit quantization (Liu et al., 2024); paged-attention serving frameworks (Kwon et al., 2023) supply the memory management underneath. Chari et al. (2025) trains a model to *preserve* uncompressed-model behavior under compression, a complementary objective to ours. Yang et al. (2024) document compression-induced hallucination and propose a representation-side fix. We use these as upstream sources of survival masks; calibrated abstention as a trained behavior is orthogonal.

**Abstention and selective generation.** Self-knowledge and selective generation in LLMs have been studied via calibration (Kadavath et al., 2022), self-knowledge benchmarks (Yin et al., 2023), verbalized confidence (Lin et al., 2022), self-evaluation gating (Ren et al., 2023), refusal-aware tuning (Zhang et al., 2024), retrieve-and-critique (Asai et al., 2024), multi-LLM collaboration (Feng et al., 2024), and the known-unknown boundary (Amayuelas et al., 2024). Kamath et al. (2020) formalize selective QA under domain shift; the closest neighbor is Joren et al. (2025) on context sufficiency in RAG. Our source of insufficiency is KV-cache eviction, and the trained behavior is bidirectional rather than gating.

## 7 Conclusion

KV-cache compression makes long-context inference more efficient, but it can also remove the very evidence needed to answer a downstream query. This paper formulates that failure mode as a behavioral learning problem: given a compression-modified context, the model should answer when the surviving evidence supports the answer and abstain when it does not. We show that this behavior can be learned with a small LoRA adapter trained from compressor masks and tight answer-bearing



spans.

These results suggest that KV-cache compression should not be evaluated only by how well it preserves uncompressed-model outputs. When compression changes what evidence is available, the model’s behavior should change as well. Compression-aware abstention offers one way to make that behavior explicit: preserve answers when the evidence survives, and refuse when compression has made the context insufficient.

## Limitations

**Aggregate accuracy is exploitable.** Because the gold label distribution is highly skewed as a function of retention ratio, aggregate honest accuracy can reward policies that do not inspect the truncated context at all. As shown in §5.2, a simple ratio-threshold rule can beat both trained adapters on aggregate by exploiting the fact that low-retention examples are mostly ABSTAIN-gold and high-retention examples are mostly CONFIDENT-gold. We therefore report aggregate numbers for orientation, but treat per-ratio and class-conditional comparisons as the more meaningful tests of compression-aware abstention. The mid-retention prompt comparison is also underpowered: at  $r = 0.50$ , the mixture-vs-prompt advantage trends positive but is not statistically significant ( $\Delta = +0.07$ ,  $p = 0.17$  pooled). Our strongest prompt-vs-training evidence comes instead from the high-retention CONFIDENT-gold slice, where prompting over-abstains and the trained adapter does not.

**The compressed-cache deployment gap.** The adapter improves substantially under actual compressed-cache decoding, but it does not close the gap to prompt-style inference. At  $r = 0.80$ , mix-cc honest accuracy under compressed-cache inference is 0.315, compared with 0.658 for mix-ps under prompt-style inference at the same retention. This should be interpreted together with the base-model collapse in the same deployment regime: under compressed-cache inference, the unaided base answers only 1/69 of CONFIDENT-gold examples correctly at  $r = 0.80$ , while the trained adapter answers 22/69. Thus cc-training yields a large relative lift over the relevant deployment baseline, but high-retention compressed-cache answering remains a real generalization limit.

**Tight-span labels are imperfect.** Our labels are based on whether short answer-bearing spans survive compression, not on a full semantic proof that the truncated context supports the answer. On a stratified 400-example audit, these labels show moderate agreement with an LLM-judge sufficiency verdict (Cohen’s  $\kappa = 0.61$ , raw agreement 80%). The main disagreement mode occurs in multi-hop questions where the answer string survives but the bridging evidence needed to connect the question to that answer does not. This means the tight-span rule is effective for constructing a training signal, but it is not exact ground truth for contextual sufficiency.

**Hard high-retention cases remain.** The adapter still fails on rare cases where the overall retention ratio is high but the specific evidence needed for the question has been evicted. At  $r = 0.80$ , across three v1 seeds, the adapter converges on hallucinating the same 4 of 7 rare ABSTAIN-gold examples. This suggests that the model has learned much of the easy version of compression-aware abstention (using retention ratio and deletion patterns as strong cues) but has not fully learned the harder behavior of introspecting on whether the specific evidence required for the current question survived.

**Scope and experimental coverage.** Our training data is limited to MuSiQue 2-hop QA, and the tight-span construction relies on `question_decomposition` annotations that are not available in all QA datasets. We replicate the main behavior on RULER multi-key NIAH and on a held-out Llama-3.1-8B-Instruct base model, but we do not retrain on another multi-hop QA dataset such as HotpotQA. Our compressed-cache evaluation also uses KVzipPress; other kvpress compressors use different eviction and position-indexing paths, and cross-compressor cc generalization remains untested. Finally, we use greedy decoding and select hyperparameters on the same validation set used for headline reporting. The search is small and a disjoint 100-example split reproduces the effect, but a fully pre-registered dev/test split and broader sampling regimes are left to future work. Finally, examples whose minimum span survival falls in the uncertain zone are excluded from both training and evaluation (Eq. 2), so the adapter is never trained on, and we never measure, the partially-compressed middle of the distribution.

## Acknowledgments

We thank the anonymous reviewers for their careful and constructive feedback. This work received no external funding, and the authors declare no competing interests.

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## A Label entropy under tight-span vs paragraph-level supervision

This is the pre-training diagnostic that motivated the tight-span pivot (§3.1). We sweep retention ratio  $r$  and compute the entropy (in nats) of the resulting three-class label distribution under two labeling schemes. *Paragraph-level*: assign CONFIDENT/ABSTAIN/DROP based on the retention of the entire supporting paragraph ( $\sim 100$  tokens). *Tight-span*: based on the retention of the  $\leq 15$ -token answer-bearing span. Both schemes use the same labeler thresholds ( $\tau_{\text{low}} = 0.3$ ,  $\tau_{\text{high}} = 0.8$ ) and the same 50 answerable MuSiQue validation rows; with three active classes the maximum attainable entropy is 1.099.

| $r$         | paragraph entropy | tight-span entropy |
|-------------|-------------------|--------------------|
| 0.05        | 0.000             | 0.000              |
| 0.10        | 0.000             | 0.098              |
| 0.20        | 0.098             | 0.471              |
| 0.30        | 0.551             | 0.803              |
| <b>0.50</b> | <b>0.000</b>      | <b>0.950</b>       |
| 0.80        | 0.686             | 0.665              |

Table 5: Label entropy (nats) at six retention ratios, on a 50-example diagnostic pool. Paragraph-level supervision collapses at the headline mid-ratio; tight-span supervision retains per-example variance there. This diagnostic was run before the main pipeline, using a small mask generator (Qwen3-0.6B) to make the sweep cheap; the production pipeline generates masks with the target base model itself (§3.2).

The headline mid-slice  $r = 0.50$  is the operative comparison. There, paragraph-level supervision has zero label entropy because every one of the 50 examples lands in DROP: at  $r = 0.50$  each  $\sim 100$ -token paragraph retains roughly half its tokens, so  $\rho$  falls in the uncertain zone  $[\tau_{\text{low}}, \tau_{\text{high}})$  for every example and the scheme yields no usable training signal at all. Tight-span supervision at the same ratio reaches 0.950, the highest value in either column, restoring the per-example variance required for Eq. 2 to be discriminative. The two schemes are closer at the extreme ratios, where retention is decisive under either granularity.

## B Output classification categories

For each (record,  $r$ ) pair we apply the compressor’s survival mask to construct  $\tilde{c}$ , run greedy decoding (max 64 new tokens), and classify the output:

- *correct\_confident* (CONFIDENT-gold, model emits the gold answer)
- *correct\_abstention* (ABSTAIN-gold, model emits the refusal string)
- *hallucinated* (ABSTAIN-gold, model emits a confident wrong answer)
- *wrong\_answer* (CONFIDENT-gold, model emits a wrong answer)
- *overcautious* (CONFIDENT-gold, model emits the refusal string)
- *lucky\_guess* (rare; tracked separately)

Honest accuracy is  $(\text{correct\_confident} + \text{correct\_abstention})/\text{total}$  at each  $r$ . Hallucination rate is  $\text{hallucinated}/\text{total}$ . Overcaution rate is  $\text{overcautious}/n_{\text{CONFIDENT-GOLD}}$ . Conditional wrong-when-answering rate is  $\text{wrong\_answer}/(\text{model\_emits\_answer\_count})$  on CONFIDENT-gold examples.

**Matching rules.** Scoring is substring-based, not exact-match. A response counts as an abstention if its first 200 characters (lower-cased) contain a refusal phrase (e.g., “i cannot determine”, “cannot determine the answer”), which captures refusal-string variants rather than only the exact training target. A response counts as containing the gold answer if the gold string or one of its answer aliases occurs as a case-insensitive substring. The category is then: for ABSTAIN-gold, *correct* if it abstains else *hallucinated*; for CONFIDENT-gold, *overcautious* if it abstains, *correct* if it contains the gold answer, else *wrong\_answer*. A response that abstains in its opening clause is scored as an abstention even if other text follows.



## C Calibration sub-checks

We split adapter behavior on CONFIDENT-gold examples into two failure modes: *overcaution* (CONFIDENT-gold answered with the refusal string) and *wrong-when-answering* (CONFIDENT-gold answered with a wrong answer). We report at the two slices with sufficient CONFIDENT-gold sample size,  $r \in \{0.50, 0.80\}$ .

| $r$  | $n_C$ | base ovc | v1 ovc | mix ovc    | v1 wrong/ans | mix wrong/ans |
|------|-------|----------|--------|------------|--------------|---------------|
| 0.50 | 22    | 23%      | 32%    | <b>23%</b> | 33%          | <b>18%</b>    |
| 0.80 | 69    | 17%      | 3%     | 10%        | 34%          | <b>27%</b>    |

Table 6: Calibration sub-check on KVzip val (seed 42). v1 preserves conditional answer quality with the +9pp  $r=0.50$  overcaution cost flagged in §5.1. Mixture closes the overcaution cost while improving conditional answer quality.

**Mixture cross-compressor calibration.** Conditional wrong-when-answering rate decreases on 7 of 8 conclusive (compressor,  $r$ ) cells under mixture training.

| $r$  | Compressor   | $n_C$ | v1 wrong/ans | mix wrong/ans | v1 ovc | mix ovc |
|------|--------------|-------|--------------|---------------|--------|---------|
| 0.50 | KVzip        | 22    | 33%          | <b>18%</b>    | 32%    | 23%     |
| 0.50 | ExpAttn      | 23    | 29%          | <b>17%</b>    | 9%     | 22%     |
| 0.50 | SnapKV (ctx) | 18    | 33%          | 40%           | 17%    | 17%     |
| 0.50 | SnapKV (nat) | 28    | 42%          | <b>35%</b>    | 14%    | 18%     |
| 0.80 | KVzip        | 69    | 34%          | <b>27%</b>    | 3%     | 10%     |
| 0.80 | ExpAttn      | 65    | 41%          | <b>28%</b>    | 3%     | 8%      |
| 0.80 | SnapKV (ctx) | 59    | 41%          | <b>27%</b>    | 0%     | 7%      |
| 0.80 | SnapKV (nat) | 67    | 42%          | <b>32%</b>    | 0%     | 6%      |

Table 7: Mixture-adapter calibration sub-check on all four val sets,  $r \in \{0.50, 0.80\}$ . Conditional wrong-when-answering improves on 7 of 8 cells.

## D Cross-compressor evidence

| $r$              | KVzip | ExpAttn | SnapKV (ctx) | SnapKV (nat) |
|------------------|-------|---------|--------------|--------------|
| 0.05             | +0.85 | +0.70   | +0.39        | +0.40        |
| 0.10             | +0.72 | +0.66   | +0.26        | +0.30        |
| 0.20             | +0.56 | +0.59   | +0.17        | +0.11        |
| 0.30             | +0.43 | +0.35   | +0.12        | +0.04        |
| 0.50             | +0.21 | +0.29   | +0.13        | +0.03        |
| 0.80             | +0.10 | +0.08   | +0.05        | +0.04        |
| <b>aggregate</b> | +0.53 | +0.49   | +0.21        | +0.20        |

Table 8: Per-ratio  $\Delta$ honest accuracy of the v1 adapter (seed 42) on four compressors. Within-family transfer (ExpAttn) tracks KVzip closely; cross-family transfer (SnapKV) is partial and most degraded at low  $r$ , where truncation-pattern divergence is greatest.

**Two-pattern decomposition.** On 10 ABSTAIN-labeled qualitative examples evaluated under all four compressors, the adapter (trained only on KVzip masks) correctly *answers* under Expected Attention on 2 examples and under SnapKV-native on 3 examples, cases where the held-out compressor happened to retain supporting tokens that KVzip evicted. The read-and-answer behavior is therefore surviving-evidence-conditional: it transfers wherever  $\tilde{c}$  contains the answer, regardless of which compressor produced the truncation. The recognize-missing-evidence behavior is truncation-pattern-specific: 96% of ExpAttn correct-abstentions, 97% of SnapKV-context correct-abstentions, and 98% of SnapKV-native correct-abstentions are also correct abstentions under KVzip: the held-out compressors do not enable abstention recognition on examples KVzip didn’t enable; they cause failure-to-recognize on examples KVzip did enable.

## E Mixture training

### E.1 Dataset

We construct a training distribution from three compressors at six retention ratios. Per-(compressor,  $r$ ) class balance is applied for  $r \geq 0.20$  at the standard 30/70 minority floor; at  $r \in \{0.05, 0.10\}$  the natural distribution is kept (CONFIDENT-gold rate  $< 1\%$  across compressors). Final mixture train: 8772 instances. Holdout (for checkpoint selection only, disjoint by example ID from training): 984. Val: 1511 instances across all three compressors.

### E.2 Seed robustness

| Compressor         | seed 42 | seed 43 | spread |
|--------------------|---------|---------|--------|
| KVzip              | +0.5378 | +0.5551 | 0.017  |
| Expected Attention | +0.5676 | +0.5676 | 0.000  |
| SnapKV (ctx)       | +0.5336 | +0.5267 | 0.007  |
| SnapKV (nat)       | +0.4272 | +0.4296 | 0.002  |

Table 9: Aggregate  $\Delta$ honest accuracy across mixture seeds. Max spread across all four compressors is 0.017.

Per-cell spread at  $r = 0.80$  on SnapKV is large (up to 0.23 with sign flips). These cells have very small ABSTAIN-gold  $n$  (4 KVzip, 1 ExpAttn, 10 SnapKV); single-example flips swing the percentage. The aggregate metric integrates over the full val and is robust to per-cell noise.

## F Held-out test split

We carve a 100-example test split from val (disjoint by example ID from a 363-example selection split, never seen during checkpoint selection) and re-evaluate the seed-42 v1 adapter without retraining. Aggregate  $\Delta$ honest accuracy: +0.63 on test vs full val’s +0.53. The aggregate lift is attributable to chance class skew (the random shuffle landed on a slice with 85% ABSTAIN-gold vs full val’s 79%). Selection split alone ( $n = 363$ ) shows +0.50, within 3pp of full val. Hallucination count on test split: 1/85 (vs base 59/85). The v1 result is not overfit to selection val.

## G Llama-3.1-8B-Instruct replication

We retrain the entire pipeline with Llama-3.1-8B-Instruct as the base and KVzip mask-generator, via the `NousResearch/Meta-Llama-3.1-8B-Instruct` mirror (identical weights and tokenizer). Same labeler thresholds, same 30/70 floor, same disjoint-by-id discipline. Llama train: 3640 instances (vs Qwen’s 2669) because Llama’s KVzip retains slightly more answer-bearing tokens at the same ratios.

| $r$              | Qwen $\Delta$ hon | Llama $\Delta$ hon | abs. diff   |
|------------------|-------------------|--------------------|-------------|
| 0.05             | +0.85             | +0.84              | 0.01        |
| 0.10             | +0.72             | +0.82              | 0.10        |
| 0.20             | +0.56             | +0.83              | 0.27        |
| 0.30             | +0.43             | +0.73              | 0.30        |
| 0.50             | +0.21             | +0.20              | 0.01        |
| 0.80             | +0.10             | +0.19              | 0.09        |
| <b>aggregate</b> | <b>+0.53</b>      | <b>+0.64</b>       | <b>0.11</b> |

Table 10: Per-ratio  $\Delta$ honest accuracy on Qwen2.5-7B (v1) vs Llama-3.1-8B (independently trained pipeline). Both monotone-decreasing; aggregate magnitude comparable.

The Llama adapter eliminates 96% of base hallucinations ( $228 \rightarrow 10$ ), preserves the monotone abstention curve (1.00/1.00/1.00/0.85/0.20/0.05 across  $r$ ), and produces +298 correct answers vs base (Qwen: +244). Differences from Qwen reflect differences in label distribution: Llama’s val gold-mix at  $r = 0.50$  is CONF-heavy (35/15) vs Qwen’s (22/26), so Llama’s lower abstention rate at 0.50 (0.20 vs Qwen’s 0.58) is correct on its slice.

## H Failure mode detail

At  $r = 0.80$ , the remaining failure is concentrated in rare ABSTAIN-gold examples. Decomposition:

- On 4 ABSTAIN-gold examples at  $r = 0.80$ , the adapter hallucinates on all 4 (vs base 3/4); slice too small ( $n = 4$ ) to draw conclusions in isolation, but the four examples replicate as the same items across seeds.
- On 69 CONFIDENT-gold examples at  $r = 0.80$ , adapter wrong-answer count is 23 vs base 21, but answer count is 67 vs base 57, so conditional wrong-when-answering *improves* from 37% to 34%.

The remaining failure mode is therefore not sloppy answering on CONFIDENT-gold (conditional answer quality actually improves under the adapter); it is the rare ABSTAIN-gold cases at high apparent retention. The adapter learned the easy version of compression-aware abstention (use  $r$  as a strong prior) and not the hard version, which would require introspecting on whether the specific tokens that mattered for this question survived.

A replication artifact strengthens the reading: across three seeds with different LoRA initializations, the adapter hallucinates on exactly 7 ABSTAIN-gold examples per seed, but only 4 of the 7 are stable across all three seeds. Two of the 4 stable cases are at  $r = 0.80$ , two at  $r = 0.50$ . The high-fill counter-instances are systematically harder; three independent training trajectories converge on the same failures.

## I Qualitative examples

Three representative ABSTAIN-labeled examples on which the base model produces confident wrong answers and the adapter abstains correctly:

**Q:** Who founded the company that distributed the film UHF?

**Gold:** Mike Medavoy

**Base:** “Kevin Smith founded the distribution company that released the film UHF...”

**Adapter:** “I cannot determine the answer to this question from the available context.”

**Q:** Which company owns the manufacturer of Learjet 60?

**Gold:** Bombardier Inc.

**Base:** “Fieldillusionist is credited with appearances in horror film media...”

**Adapter:** “I cannot determine the answer to this question from the available context.”

**Q:** What instrument is played by the person from The Blackout All-Stars?

**Gold:** conga

**Base:** “... specifically mentioned to be a Chapman Stick, which is a ten- or twelve-stringed instrument...”

**Adapter:** “I cannot determine the answer to this question from the available context.”

In each case, the base model picks up distractor material from  $\tilde{c}$  and constructs a confident wrong answer. The adapter, given the same input, declines.

## J Pre-training fragment-recoverability check

Before training, we performed a sanity check on 30 ABSTAIN-labeled examples at  $r = 0.30$ . We extracted the kept tokens via the survival mask, decoded them to text, and presented the resulting fragment as a fresh prompt to a clean Qwen2.5-7B-Instruct. As with the sweep in Appendix A, this pre-training diagnostic generates masks with a small KVzip pass (Qwen3-0.6B) rather than the production base model. The model recovered the correct answer in 0 of 30 cases. In approximately one third of cases the model produced confident, specific wrong answers grounded in distractor material that had survived eviction, the same failure mode the adapter is intended to fix. The check rules out the alternative explanation that ABSTAIN labels are too aggressive for the underlying inputs.

## K Baseline comparison statistics

We report bootstrap CIs and paired McNemar exact tests for the  $r = 0.50$  slice across all four compressors and at high retention ( $r = 0.80$ ) on CONFIDENT-gold examples. All bootstrap operations use seed 42 with 10,000 resamples; McNemar tests are exact (two-sided binomial on the discordant pair counts). Pooled rows aggregate the discordant counts across compressors and thus treat each (example, compressor) pair as independent; because the same example ids recur across compressors, the pooled  $p$ -values are descriptive and the per-compressor tests are primary.

| Comparison (at $r = 0.50$ )     | Compressor | $b / c$ | exact $p$             | $\Delta$ [95% CI]       |
|---------------------------------|------------|---------|-----------------------|-------------------------|
| mix vs P2                       | KVzip      | 12 / 5  | 0.143                 | +0.15 [-0.02, +0.31]    |
| mix vs P2                       | ExpAttn    | 10 / 5  | 0.302                 | +0.12 [-0.07, +0.29]    |
| mix vs P2                       | SnapKV-ctx | 5 / 5   | 1.000                 | +0.00 [-0.15, +0.15]    |
| mix vs P2                       | SnapKV-nat | 5 / 6   | 1.000                 | -0.03 [-0.19, +0.14]    |
| mix vs P2 (pooled)              | all four   | 32 / 21 | 0.169                 | +0.066 [-0.024, +0.151] |
| mix vs Frag                     | KVzip      | 16 / 4  | 0.012                 | +0.25 [+0.08, +0.42]    |
| mix vs Frag                     | ExpAttn    | 15 / 1  | 0.0005                | +0.33 [+0.17, +0.50]    |
| mix vs Frag                     | SnapKV-ctx | 16 / 4  | 0.012                 | +0.31 [+0.10, +0.51]    |
| mix vs Frag (pooled)            | all four   | 56 / 13 | $1.7 \times 10^{-7}$  | +0.259 [+0.169, +0.349] |
| at $r = 0.80$ on CONFIDENT-gold |            |         |                       |                         |
| mix vs P2                       | KVzip      | 25 / 3  | $2.7 \times 10^{-5}$  | +0.32 [+0.19, +0.45]    |
| mix vs P2                       | ExpAttn    | 25 / 4  | $1.0 \times 10^{-4}$  | +0.32 [+0.18, +0.46]    |
| mix vs P2                       | SnapKV-ctx | 23 / 3  | $8.8 \times 10^{-5}$  | +0.34 [+0.19, +0.47]    |
| mix vs P2                       | SnapKV-nat | 22 / 3  | $1.6 \times 10^{-4}$  | +0.28 [+0.15, +0.42]    |
| mix vs P2 (pooled)              | all four   | 95 / 13 | $1.5 \times 10^{-16}$ | +0.32 [+0.25, +0.38]    |

Table 11: Paired McNemar exact tests and paired-bootstrap 95% CIs. “mix vs P2” is mixture adapter vs prompt-only abstention baseline; “mix vs Frag” is mixture adapter vs base on the same compression-truncated input. The mid-slice prompt comparison does not reach significance even pooled; the high-retention prompt comparison is decisive on every compressor and pooled.

## L Answerability preservation

For each example with `gold = CONFIDENT`, we compute whether the adapter answers correctly (`classification = correct` and `model_abstained = False`). The preservation rate is this count divided by the total CONFIDENT-gold count; the preservation ratio is adapter-rate divided by base-rate. A ratio  $\geq 1.0$  means the adapter answers correctly at-least-as-often as base.

## M Compressed-cache training details

**Training under cc decoding.** For each (context, query, target) training tuple, we wrap the forward pass in a `kvpress` context manager. Prefill processes the full context; the `press` evicts low-scoring K/V pairs from the cache; cross-entropy loss is computed on target tokens decoded from the post-eviction cache state. The compression algorithm at training time matches each example’s source compressor (`compressor` field in the mixture training data).

We use the mixture training set ( $\sim 8772$  instances across KVzip, ExpAttn, SnapKV-context). Hyperparameters match v1 (LoRA rank 16,  $\alpha = 32$ , dropout 0.05; AdamW; constant LR; bf16; FlashAttention-2; max seq len 4096; batch size 1, gradient accumulation 16; 3 epochs). We sweep  $\text{lr} \in \{1 \times 10^{-4}, 2 \times 10^{-4}\}$  on seed 42 and select the better LR by val loss with sub-epoch checkpointing every 25% of an epoch and a fluency spot-check at each saved checkpoint (responses on five  $r = 0.80$  CONFIDENT-gold val examples must be fluent, not sub-token gibberish). The selected configuration was used to train seed 43; seed 44 was attempted but did not complete.

**Per-seed mix-cc results.** Aggregate honest accuracy under KVzipPress at inference: seed 42 = 0.821, seed 43 = 0.827, mean = 0.824, two-seed spread 0.006. Per-ratio shape preserved across seeds:  $r = 0.05$ : 1.000/1.000;  $r = 0.10$ : 0.989/0.989;  $r = 0.20$ : 0.963/0.976;  $r = 0.30$ : 0.923/0.954;  $r = 0.50$ : 0.500/0.458;  $r = 0.80$ : 0.315/0.342.

**Single-compressor cc-training.** We also trained a cc-trained adapter on KVzip masks alone (analogous to v1 but under cc decoding). Aggregate honest accuracy under cc inference: 0.734, statistically tied with mix-ps cc (0.730); but at  $r = 0.80$  on CONFIDENT-gold, the single-compressor cc-trained adapter reduces over-abstention to 0.203 while increasing the wrong-answer count (confident wrong answers on CONFIDENT-gold, not the ABSTAIN-gold hallucinations of Appendix B) to 42/69 (61%). Multi-compressor cc-training (mix-cc) yields a wrong-answer count of 21/69 (30%). Qualitative inspection of the single-compressor adapter’s high-retention failures shows sub-token-fragment outputs (“Acfaairs”,



| Condition                                                | $n_C$ | base rate [CI]     | adapter rate [CI]  | ratio |
|----------------------------------------------------------|-------|--------------------|--------------------|-------|
| Aggregate (all retention ratios), prompt-style inference |       |                    |                    |       |
| v1 ps, KVzip                                             | 97    | 0.485 [0.39, 0.58] | 0.557 [0.46, 0.65] | 1.15  |
| v1 ps, ExpAttn                                           | 94    | 0.468 [0.37, 0.57] | 0.574 [0.47, 0.67] | 1.23  |
| v1 ps, SnapKV-ctx                                        | 88    | 0.545 [0.44, 0.65] | 0.568 [0.46, 0.67] | 1.04  |
| v1 ps, SnapKV-nat                                        | 127   | 0.567 [0.48, 0.65] | 0.575 [0.49, 0.66] | 1.01  |
| mix ps, KVzip                                            | 97    | 0.485 [0.39, 0.58] | 0.608 [0.51, 0.70] | 1.25  |
| mix ps, ExpAttn                                          | 94    | 0.468 [0.37, 0.57] | 0.638 [0.54, 0.73] | 1.36  |
| mix ps, SnapKV-ctx                                       | 88    | 0.545 [0.44, 0.65] | 0.591 [0.49, 0.69] | 1.08  |
| mix ps, SnapKV-nat                                       | 127   | 0.567 [0.48, 0.65] | 0.567 [0.48, 0.65] | 1.00  |
| Restricted to $r = 0.80$ slice, prompt-style inference   |       |                    |                    |       |
| v1 ps, KVzip                                             | 69    | 0.522 [0.41, 0.64] | 0.638 [0.52, 0.74] | 1.22  |
| mix ps, KVzip                                            | 69    | 0.522 [0.41, 0.64] | 0.652 [0.53, 0.75] | 1.25  |
| mix ps, ExpAttn                                          | 65    | 0.492 [0.38, 0.61] | 0.662 [0.54, 0.77] | 1.34  |
| mix ps, SnapKV-ctx                                       | 59    | 0.542 [0.42, 0.66] | 0.678 [0.55, 0.78] | 1.25  |
| Compressed-cache inference (KVzipPress)                  |       |                    |                    |       |
| mix ps cc, KVzip                                         | 97    | 0.041 [0.02, 0.10] | 0.258 [0.18, 0.35] | 6.25  |
| mix cc cc, KVzip                                         | 97    | 0.041 [0.02, 0.10] | 0.247 [0.17, 0.34] | 6.00  |
| mix ps cc, KVzip ( $r=0.80$ )                            | 69    | 0.014 [0.00, 0.08] | 0.290 [0.20, 0.41] | 20.0  |
| mix cc cc, KVzip ( $r=0.80$ )                            | 69    | 0.014 [0.00, 0.08] | 0.319 [0.22, 0.44] | 22.0  |

Table 12: Answerability preservation rates. Wilson 95% CIs. Under prompt-style inference, all 12 cells have preservation ratio  $\geq 1.0$ : the adapter does not sacrifice correct answers. Under compressed-cache inference, base collapses on CONFIDENT-gold (4% aggregate, 1.4% at  $r=0.80$ ); adapter rates of 25–32% (raw 4/97  $\rightarrow$  25/97 aggregate, 1/69  $\rightarrow$  22/69 at  $r=0.80$ ) represent a 6–22 $\times$  relative lift, with non-overlapping Wilson CIs.

“750pxI came, I I loaves and two fish.”) consistent with the cross-entropy loss over-fitting to short outputs without learning to extract content from kept tokens; the multi-compressor mix-cc residual wrong answers are cleaner pattern collisions (“March 44 BC”  $\rightarrow$  “28 November 1912”).

## N LLM-judge label validation

To probe the validity of the tight-span retention label as a proxy for answerability, we sample 400 labeled examples from `train_v1.jsonl` (stratified: 200 CONFIDENT, 200 ABSTAIN; balanced across retention ratios; sample seed 42), and query a strong judge model (GPT-5, reasoning effort low, seed 42) for a YES/NO/PARTIAL sufficiency verdict given the truncated context, query, and gold answer. The judge does not see our label or the full context.

| Statistic                                           | Value      |
|-----------------------------------------------------|------------|
| Total sampled                                       | 400        |
| Judge PARTIAL count                                 | 48 (12.0%) |
| $\kappa$ (strict, excluding PARTIAL, $n = 352$ )    | 0.609      |
| $\kappa$ (lenient, PARTIAL $\rightarrow$ CONFIDENT) | 0.525      |
| $\kappa$ (lenient, PARTIAL $\rightarrow$ ABSTAIN)   | 0.545      |

Table 13: Cohen’s  $\kappa$  between tight-span labels and LLM-judge sufficiency verdicts. Moderate agreement (Landis-Koch range 0.4–0.7).

The confusion matrix shows directional asymmetry: of 200 CONFIDENT labels, 123 are confirmed YES, 55 rejected as NO, 22 marked PARTIAL; of 200 ABSTAIN labels, 160 confirmed NO, 14 rejected as YES, 26 PARTIAL. Per-retention agreement is near-perfect at extreme low retention ( $r \in \{0.05, 0.10\}$ : 0.97–1.00) and 0.69–0.76 from  $r = 0.20$  upward. The dominant disagreement pattern (manually inspected on 25 and 13 sampled cases from the two disagreement directions) is bridge-evidence loss in multi-hop questions: the answer string survives in the kept tokens, and one of the two reasoning hops survives, but the bridging evidence linking the hops does not. The tight-span rule cannot distinguish “answer string is in the kept tokens” from “the reasoning chain that links the question to the answer is recoverable”: a known limitation of substring-based answerability surrogates that affects mid-retention multi-hop QA most.

## O Mechanism: controlled-deletion experiments

Three controlled-deletion experiments probe whether the trained adapter uses evidence content or input-length priors.

**10A: Fixed retained length.** Hold input length constant at  $L = 500$  kept tokens (selected by KVzip rank in context order) regardless of the original ratio. On the 208 examples that have  $\geq L$  kept tokens at any ratio:

| Adapter | Condition  | CONF hon [CI]      | ABS hon [CI]       |
|---------|------------|--------------------|--------------------|
| mix-ps  | fixed- $L$ | 0.147 [0.09, 0.23] | 0.885 [0.81, 0.93] |
| mix-ps  | orig- $r$  | 0.621 [0.52, 0.71] | 0.823 [0.74, 0.88] |
| mix-cc  | fixed- $L$ | 0.063 [0.03, 0.13] | 0.956 [0.90, 0.98] |
| mix-cc  | orig- $r$  | 0.274 [0.19, 0.37] | 0.956 [0.90, 0.98] |

Table 14: Test 10A. Holding input length to 500 reduces CONFIDENT-gold accuracy ( $-47\text{pp}$  mix-ps,  $-21\text{pp}$  mix-cc) but not ABSTAIN-gold accuracy. The drop is consistent with both length-prior and evidence-loss explanations; tests 10B and 10C disambiguate.

**10B: Adversarial tight-span eviction.** Construct a mask = (existing kept-mask) MINUS (tight-span token positions identified from `supporting_spans`) PLUS (random non-kept positions to refill to original length). Same length, same retention count, but answer-bearing tokens forcibly removed.

Of 97 CONFIDENT-gold examples (all retention ratios): mix-ps abstention rises from 16.5% on the original mask to 56.7% on the adversarial mask ( $+40.2\text{pp}$ ); mix-cc rises from 64.9% to 88.7% ( $+23.8\text{pp}$ ). Per-pair, 41/97 (mix-ps) and 25/97 (mix-cc) flip from “answered on original” to “abstained on adversarial.” The adapter responds to evidence content, not just length.

**10C: Opposite-priority pairs.** For each of 22 CONFIDENT-gold examples at  $r = 0.50$ , construct two masks of identical length: Mask A keeps KVzip’s top-50% (tight spans likely retained), Mask B keeps KVzip’s bottom-50% (tight spans likely evicted). On Mask A, mix-ps answers correctly 14/22; on Mask B, 0/22 correct, 22/22 abstentions. mix-cc: A 5/22, B 0/22 correct with 20/22 abstentions. At identical input length and identical example, the adapter’s answer-vs-abstain decision flips entirely based on which half of the context is kept. Neither adapter answers correctly on Mask B; mix-ps abstains on all 22, and mix-cc’s two non-abstentions are confident wrong answers.

## P Random-mask training-signal probe

To probe whether the contribution depends on the importance-weighted nature of compressor masks or on the abstention objective itself, we retrain v1 with masks drawn uniformly at random at matched per-example retention counts, holding everything else constant. Under prompt-style inference, the random-mask adapter matches the mixture adapter on KVzip val (aggregate 0.873 vs mixture 0.86; paired McNemar  $\Delta = +0.017$ ,  $p = 0.23$ ) and beats v1 on cross-family transfer ( $+0.276$  on SnapKV-context,  $p \approx 10^{-24}$ ). Two implications: under ps inference the abstention objective is the load-bearing signal, importance-weighted mask structure is not required to match v1’s headline numbers; and single-compressor training induces an anti-transfer signature: v1 underperforms even random-mask training on a held-out compressor. Both findings motivate compressor-mixture training; §5.5 shows multi-compressor training delivers further gains under cc inference that single-compressor training (random or KVzip-only) does not.

## Q Long-context generalization (RULER)

We evaluate the v1 adapter (no retraining) on RULER multi-key NIAH (Hsieh et al., 2024) (complementary to LongBench (Bai et al., 2024)) at three context lengths up to 16K, four times the training maximum. At 16K, abstention rate on ABSTAIN-gold across  $r$  is 67%/76%/74%/29%/11%/0%: the same broadly declining shape as MuSiQue val, though not strictly monotone at the two lowest ratios (Figure 5a).

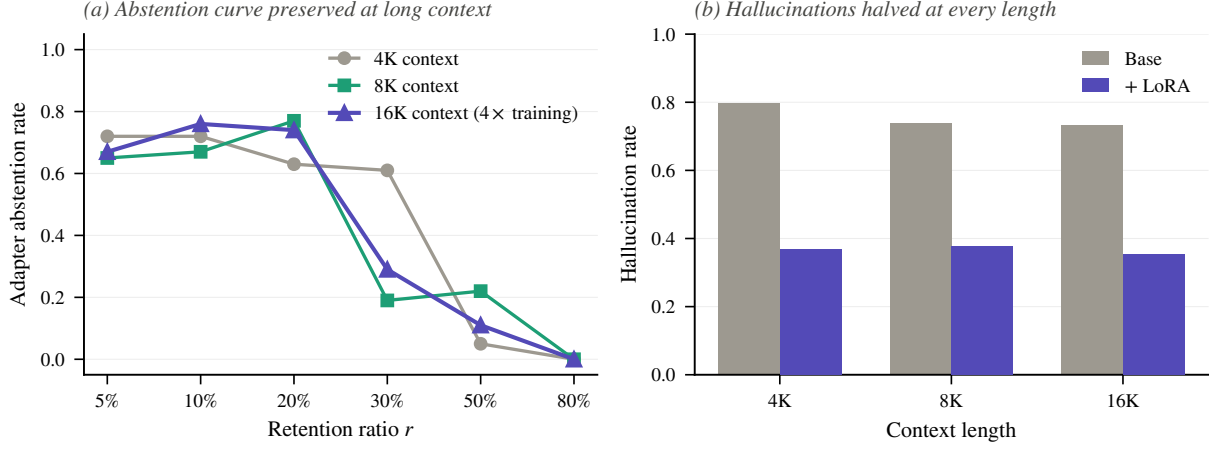


Figure 5: (a) Adapter abstenion rate on ABSTAIN-gold at three context lengths on RULER multi-key NIAH; the monotone-decreasing-with- $r$  shape is preserved at 4x training context. (b) Aggregate hallucination rate, base vs adapter, by length.

Aggregate hallucination rate drops by 36–43pp at every length (Figure 5b); the adapter trades hallucinations on evicted-needle examples for refusals.

## R Training details

The LoRA adapter (rank 16,  $\alpha = 32$ , dropout 0.05) adapts only the four attention projections ( $q\_proj$ ,  $k\_proj$ ,  $v\_proj$ ,  $o\_proj$ ); MLP modules are unadapted, yielding 10.1M trainable parameters (0.13% of Qwen2.5-7B-Instruct’s 7.6B). We train with AdamW at  $lr \in \{5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}\}$  (1-epoch sweep, picked by validation loss), constant LR, batch size 1 with gradient accumulation 16, max sequence length 4096, bf16 with FlashAttention-2 (Dao, 2024).